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CSA0504-DBMS

Experiment – 17

High Level Programming Extensions – Function

```
MySQL 8.0 Command Line Client
+-----+
| Tables_in_proceduredb |
+-----+
| student                |
+-----+
1 row in set (0.05 sec)

mysql> SELECT * FROM Student;
+-----+-----+-----+-----+
| StudentID | StudentName | Department | Marks |
+-----+-----+-----+-----+
| 101       | Indhu       | CSE        | 85     |
| 102       | Meghana     | IT         | 78     |
| 103       | Shalini     | ECE        | 92     |
| 104       | Teju        | CSE        | 67     |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION GetGrade(marks INT)
  -> RETURNS VARCHAR(10)
  -> DETERMINISTIC
  -> BEGIN
  ->   DECLARE grade VARCHAR(10);
  ->
  ->   IF marks >= 90 THEN
  ->     SET grade = 'A+';
  ->   ELSEIF marks >= 80 THEN
  ->     SET grade = 'A';
  ->   ELSEIF marks >= 70 THEN
  ->     SET grade = 'B';
  ->   ELSEIF marks >= 60 THEN
  ->     SET grade = 'C';
  ->   ELSE
  ->     SET grade = 'F';
  ->   END IF;
  ->
  ->   RETURN grade;
  -> END //
Query OK, 0 rows affected (0.06 sec)
```

```
mysql>
mysql> DELIMITER ;
mysql> SELECT StudentID, StudentName, Marks, GetGrade(Marks) AS Grade
-> FROM Student;
```

StudentID	StudentName	Marks	Grade
101	Indhu	85	A
102	Meghana	78	B
103	Shalini	92	A+
104	Teju	67	C

4 rows in set (0.05 sec)

```
mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION AddBonus(marks INT)
-> RETURNS INT
-> DETERMINISTIC
-> BEGIN
->     RETURN marks + 10;
-> END //
```

Query OK, 0 rows affected (0.06 sec)

```
mysql>
mysql> DELIMITER ;
mysql> SELECT StudentName, Marks, AddBonus(Marks) AS FinalMarks
-> FROM Student;
```

StudentName	Marks	FinalMarks
Indhu	85	95
Meghana	78	88
Shalini	92	102
Teju	67	77

4 rows in set (0.00 sec)

```
mysql> SHOW FUNCTION STATUS WHERE Db = 'ProcedureDB';
```

Db	Name	Type	Definer	Modified	Created	Security_type	Comment	character_set_client	collation_connection	Database Collation
proceduredb	AddBonus	FUNCTION	root@localhost	2026-08-11 11:40:15	2026-08-11 11:40:15	DEFINER		cp850	cp850_general_ci	utf8mb4_0900_ai_ci
proceduredb	GetGrade	FUNCTION	root@localhost	2026-08-11 11:39:03	2026-08-11 11:39:03	DEFINER		cp850	cp850_general_ci	utf8mb4_0900_ai_ci

2 rows in set (0.00 sec)

```
mysql> DROP FUNCTION GetGrade;
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> DROP FUNCTION AddBonus;
Query OK, 0 rows affected (0.05 sec)
```